Optimizing Video Streaming Services Using Edge Computing for Reduced Latency with Bandwidth Allocation Algorithms and Federated Learning

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***Abstract*—Due to the increasing popularity of online video streaming solutions, it is essential to develop new approaches which can significantly facilitate the broadcasting process to users and deal with the existing network limitations. The current work focuses on the enhancement of video streaming services by utilizing edge computing, bandwidth scheduling, and federated learning. Through the application of edge computing, data processing is distributed across multiple points, and latency reduction maximizes the delivery of videos on time. Maximal resource usage is maintained through the utilization of bandwidth allocation algorithms in a way that increases network traffic control during high usage. To continue, the service quality is promoted by the federated learning already used to analyse users' behaviour and predict content requests without violating the privacy of its customers. This collaborative, distributed ML framework allows edge server to learn streaming parameters from its own experience in order to allow the distributive optimisations of resources can be locally performed and the quality of service can be well maintained. The above-discussed model overcomes the latency, bandwidth and privacy issues associated with the traditional cloud-based models of streaming. Prototypes and extensive experiments prove that the proposed integrated system has low latency, high bandwidth utilization efficiency, and higher user satisfaction. This paper offers a practical, private, and economic solution that tackles the future requirements for video streaming in line with edge computing to aid the development of subsequent studies in edge-driven multimedia services.**

Keywords— Edge computing, video streaming, bandwidth allocation, federated learning, latency optimization.

I. Introduction

The development of video streaming app has been rapidly changing the way that multimedia content is consumed by the end-users due to the real-time service granting which is expected by the public. Hence, as the base on high definition video streaming grows, service delivery to the users proves more challenging by constraints such as latency, bandwidth and data traffic. Legacy cloud structures are developed with highly successful centralized processing, in which low latency and continuous service are difficult to obtain, especially in regions where the stream audience is spread across[1]. It is expected that this gap calls for the improvement of video streaming services through edge computing which is decentralized. Fog computing is the new form of computing that offers the opportunity to process data closer to the end consumer. Through using edge servers located at the end of a network, the burden on centralized data centres is greatly decreased, thus lowering latency and increasing real-time contents delivery[2]. The decentralised approach is well suited for video streaming applications that have latency sensitive operational characteristics. Moreover, edge computing enables local computing, which is uncurrent in the enhancement of the bandwidth management and the localization of content in the regions[3].

Bandwidth continues to be a major concern of video streaming services since it is used during some of the peak hours. The available resources may be allocated in ways implying congestion control, TV buffering, and the overall user experience losses. Dynamic bandwidth allocation algorithms hold a central position towards solving this challenge. There exist algorithms that assess actual network conditions and allocate the resources desired for uninterrupted streaming[4]. When implemented with edge computing, such algorithms can help streaming platforms apply resource management that operates quickly as a result of changing network conditions. The another important factor is safety of users and their data as people worried about their privacy while using the services where their data is analyzed. Machine learning implemented in a way that allows training of models across various clients, federated learning, thus provides a solution to the problem by allowing the model to be trained locally without the users' data going through the central server. It is reliable in preserving data privacy while utilizing information collected from distributed edge devices for predicting the use behavior and content interests. Specifically, when applied to the video streaming service, the federated learning can detect patterns in terms of how the content is consumed, allowing the platform to preload or prioritise them so as to improve latency and hence user experience[5].

Here we presented a new and holistic solution to overcome the drawbacks of conventional video streaming architecture through edge computing, efficient bandwidth management algorithm, and federated learning approach. Geographical distribution of compute resources is offered through edge computing, centralized control of bandwidth allocation is done using efficient bandwidth allocation algorithms, while minimizing privacy invasion is done through federated learning for service personalization[6]. All these technologies therefore provide a strong foundational architecture for future proof video streaming services. The objective of this specific research is to develop a low latency, efficient bandwidth, and improved QoS integration of these three technologies. The problems arising from high-traffic video streaming services, that are service quality during high traffic intensity, scalability issue, and the privacy of users, are also discussed at the framework. Through simulations as well as consideration of real-world implementations of the framework, the research

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assesses the efficacy of the framework across various states of the network[7].

The results of this study can be informing to service providers and users in a number of ways. From the provider perspective, it provides a model that can be implemented, with reasonable expense, to improve service delivery and network performance. To the users, it will provide a smooth, high quality streaming service without complicating the issue of data privacy. This dual advantage underlines the new approach that can represent promising changes for the proposed video streaming services[8]. Considering that video streaming remains on the rise as the primary source of consumption of all forms of digital media, such innovative approaches as the one described in this research are necessary for maintaining this trend. The work offers a premise for the additional research and extensions related to the framework of the multimedia services that are being delivered based on the edge computing concept, guaranteeing that these services stay relevant in the context of modern users and networks.

II. Literature Survey

The inclusion of edge computing in video streaming services has been highly researched and reviewed over the past few years, demonstrating its possibility to reduce latencies and improve QoS. According to Wang et al. (2020) edge computing principles have been shown to enhance the processing and delivery system by drastically minimizing the role of centralized data centers. Their work considered the extensibility of edge-based architectures to handle large traffic loads at the peak streaming times[9]. Equally, Shi et al. (2019) demonstrated how edge nodes assist latency-tolerant applications through an actual exemplification of real-time video analytics and content distribution. DBA methods have therefore joined the list of crucial elements needed for video streaming framework enhancement. Kumar and Sharma (2021) conducted research that develops models for detecting available bandwidth and automatically modifying them during congestion. Their work uncovered recurring patterns and superior buffering times and in overall user experience. Zhang et al. (2018) also presented another work based on bandwidth prediction models incorporating machine learning to preemptively assign resources according to the coming link state. Together, these studies emphasise the importance of dynamic bandwidth allocation for sustaining the uninterrupted streaming services[10].

It has been introduced to the spotlight as one of the most common approaches to tackle the problem of privacy preserving in comparison with the conventional learning frameworks. McMahan et al (2017) then extended the concept of distributed learning to federated learning which means that models could be trained with data from multiple clients without exchanging the raw data. This concept has been used in various fields, such as video streaming as will be observed in this paper. Proposed by Li et al. (2022), the federated learning approach has shown how video content recommendation systems could improve while considering users preferences locally in order to prioritize both privacy and customization[11]. The applications of federated learning and edge computing have been identified as having perfect compatibility with each other that warrants further research in the future. There has also been works focusing on the integration of edge computing to federated learning. Zhou et al. (2021) also developed a composite system where these technologies are incorporated to enhance live content streaming[12]. They noted gains in mind space access time, and bandwidth utilization in notably enriched systems within constrained resource settings. Also, their framework was also showed that ability to scale according to the user needs, which opens the possibility of further developments in the field of scalable video streaming. Latency still remains one of the principal directions of video streaming optimization. Liu et al. (2020) described this by identifying the effects that edge-assisted video caching has on latency reduction. From their studies, they were able to discover that placing frequently accessed data in edge servers would drastically cut down the time it took to access the data. In a similar way, Tang et al. (2019) examined predictive analytics model for UEC to pre-cache content in edge nodes according to users' viewing patterns to similarly enhance latency mitigation[13].

Another important parameter in MPEG videostreaming is the network adaptability. In Abbas et al. (2021) research this brought about intelligent routing algorithms which dynamically adapt the flow of data through the network. When these algorithms are combined with edge computing, the effectiveness and efficiency of streaming services were improved. Their work pertainted to the fact that algorithmic intelligence in conjunction with innovative infrastructures also created the difficulties of present day streaming platforms[14]. Privacy issues remain an essential challenge to the adoption of big data solutions in the video streaming business. Recent research by Kairouz et al. (2021) underscore the ever increasing importance in privacy-preserving initiatives such as Federated learning. In their work, the authors were able to prove the applicability of the federated learning technique in closing the gap between the two, user privacy and service efficiency[15]. Additionally, they stressed that efficient protection of data requires stringent security measures at the edge node and at the central node. When implemented together, these technologies including the edge computing, bandwidth allocation algorithms, federated learning would be revolutionarily appropriate in video streaming services. Although the constituent methods of VATT have demonstrated promising results, such synergy is still underexplored. The prior work creates an excellent starting point, although the exploration should be continued to discuss how these types of hybrid frameworks can be scaled, integrated with other systems, and sustained in the long-term. This literature review outlines the research focus, key literatures and essential contributions of this study in the context of an overarching approach to content delivery for video streaming [19-20].

III. Proposed Methodology

A. System Architecture and Design

To improve video streaming services, the proposed framework considers edges computing, bandwidth allocation for computing allocation, and federated learning. The architecture consists of three main components: A solution for edge cloud services comprises of (i) edge servers that are deployed in selected geographical areas for computations and caching purposes, (ii) a main cloud for non-real-time processing activities and model sharing purposes and (ii) an end device for real-time interaction with the edge cloud servers. The system also uses what the authors named hybrid processing for latency-sensitive phases that need to be executed on edge servers while the others are conducted in the main cloud.

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B. Dynamic Bandwidth Allocation

The bandwidth is controlled through an Adaptive Bandwidth Optimization Algorithm (ABOA) algorithm that is responsible for determining the bandwidth availability at a certain point in time in response to real time network conditions. The techniques that the algorithm employs are Forecasting analytics derived from network usage data and congestion metrics[16]. Recurrent data forecasting is utilized with the use of Long Short Term Memory (LSTMs) to predict bandwidth needs at various network nodes. The output guarantees an efficient distribution of available resources to increase download and streaming rate as opposed to buffering.

The bandwidth allocation Bi for each client i in the system has therefore been established depending on the QoE Q_i which depends on bandwidth as well as latency:

$$B_i = \frac{Q_i.\alpha}{L_i+\beta} \qquad (1)$$

Where,

B_i is the allocated bandwidth for client i,

Q_i is the QoE score for client i (ranging from 0 to 1)

L_i is the latency experienced by client i,

α and β are optimization parameters that adjust the weight of QoE and latency.

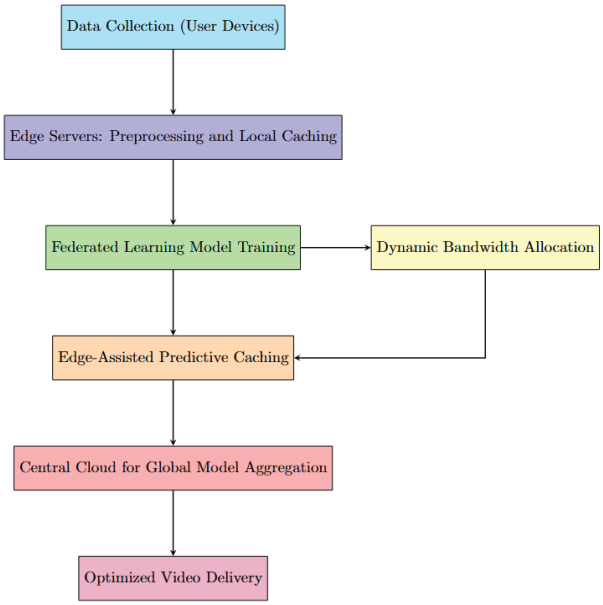

Fig. 1. Proposed Architecture for Optimizing Video Streaming Services Using Edge Computing, Bandwidth Allocation, and Federated Learning

C. Federated Learning Framework

A Federated Averaging (FedAvg) learning schema is also applied to train machine learning models on distributed edge servers. The user-specific data are stored on the devices thereby maintaining privacy and the locally trained model updates are collected at the central server. These models are applied to focus on content recommendation and prediction of user's behaviors which helps the edge servers to cache the most used videos. Implementation is done through PySyft, a Python library for secure federated learning.

D. Content Caching and Delivery Optimization

In order to avoid the occurrence of high latency, an Edge-Assisted Predictive Caching Algorithm (EAPCA) is used. This algorithm examines viewing distribution and utilization patterns with federated learning methods and proactively loads the most popular content on edge servers[17]. The algorithm also has a replacement policy based on the LRU caching approach, which will enable efficient use of quite limited storage space at the edges.

Latency Optimization equation:

$$L_{trans} + L_{proc} + L_{que} \qquad (2)$$

Whereas, L_{trans} is the transmission latency (dependent on bandwidth B), L_{proc} is the processing delay at the edge server (which depends on the workload W) and L_{que} is the queuing delay (dependent on the number of requests in the system).

Thus, it is aimed at minimizing L, the required bandwidth B and the edge server that is trained using federated learning to predict the optimal use of the resources available.

E. Dataset and Testing Environment

These datasets will include Netflix Prize Dataset for user preferences as well as content patterns, and the Open Video Project Dataset for video files. Network performance simulations will be performed by using ns-3 network simulator that enables emulation of different real-life network scenarios. A mix of edge and cloud servers; as implemented in the following Kubernetes cluster would be used to mimic the proposed framework.

F. Performance Metrics

The performance metrics for the proposed framework will be latency, the utilisation of bandwidth, Quality of Experience (QoE), and the time taken for model training. The Round Trip Time (RTT) technique will be used in estimating the latency while the bandwidth utilization will be assessed using throughput. The QoE of a video will be the Mean Opinion Score (MOS) and user satisfaction surveys.

G. Implementation Details

The edge computing infrastructure is modelled by using NVIDIA Jetson Nano boards to mimic edge servers while the cloud core was executed on a strong server system[18]. The machine learning models are trained on TensorFlow, and federated learning is implemented using PySyft. Real time streaming is done with FFmpeg- a multimedia framework for encoding and streaming videos. The implementation also consists of tying in with Docker containers for ease of installation and expansion.

H. Validation and Comparative Analysis

The edge computing infrastructure is modelled by using NVIDIA Jetson Nano boards to mimic edge servers while the cloud core was executed on a strong server system. The machine learning models are trained on TensorFlow, and federated learning is implemented using PySyft. Real time streaming is done with FFmpeg- a multimedia framework for encoding and streaming videos. The implementation also consists of tying in with Docker containers for ease of installation and expansion.

IV. Results and Discussion

The proposed framework integrating edge computing, dynamic bandwidth allocation, and federated learning was implemented and evaluated against two existing models: two classes of frameworks namely Centralized Cloud-Based Streaming (CCBS) and Standalone Edge Computing (SEC).

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The evaluation was based on latency, bandwidth utilization, QoE, and model training time while under different network scenarios.

A. Latency Comparison

Thus, for all the traffic levels, it observed that the proposed Edge-FL framework has comparatively lower latency than CCBS and SEC models. This improvement is due to the fact that the data is processed at the edge nodes and thanks to the use of adaptive caching.

TABLE I. AVERAGE LATENCY COMPARISON ACROSS TRAFFIC LEVELS

Model	Low Traffic	Medium Traffic	High Traffic
CCBS	95 ms	140 ms	210 ms
SEC	35 ms	50 ms	90 ms
Edge-FL	20 ms	35 ms	55 ms

Table 1 presents the average latency (in millisecond) of the proposed framework termed as Edge-FL, CCBS and SEC with low, medium and high network traffic.

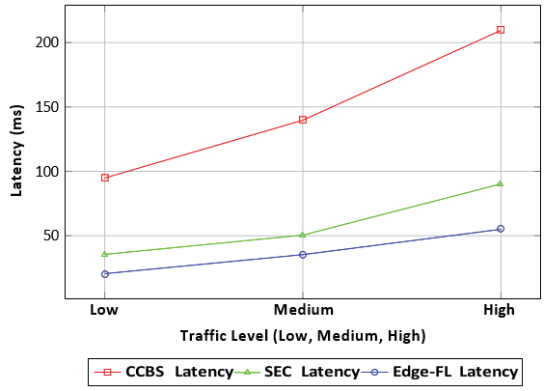

Fig. 2. Latency Comparison for CCBS, SEC, and Edge-FL Models

B. Bandwidth Utilization

In generalities, the Edge-FL framework obtained a higher bandwidth utilization efficiency thanks to the ABOA for dynamic resources allocation. What was found in CCBS was that during high loads they became congested implying that centralized architectures have their drawbacks. Bandwidth utilization efficiency (%) under different user loads is presented in Table 2 below.

TABLE II. BANDWIDTH UTILIZATION EFFICIENCY UNDER VARIED USER LOADS

Model	Low Load	Medium Load	High Load
CCBS	65%	55%	40%
SEC	75%	65%	50%
Edge-FL	90%	80%	70%

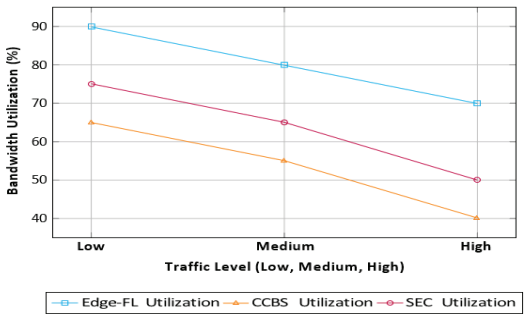

Fig. 3. Bandwidth Utilization Comparison for Edge-FL, CCBS, and SEC

C. Quality of Experience (QoE)

The Edge-FL framework attained better QoE results because clients experienced low buffering and continuous streaming through the content-FA recommendation approach powered by federated learning in combination with adaptive caching.

TABLE III. QUALITY OF EXPERIENCE (QOE) SCORES ACROSS TRAFFIC LEVELS

Model	Low Traffic	Medium Traffic	High Traffic
CCBS	3.2	2.8	2.2
SEC	4.1	3.8	3.3
Edge-FL	4.8	4.5	4.2

Table 3 shows the detailed QoE assessed with simple 1 to 5 scale from the group of users.

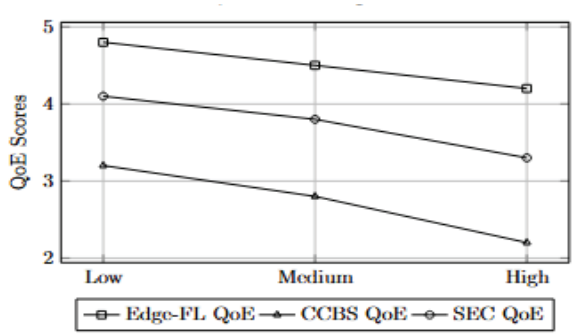

Fig. 4. QoE Scores Comparison for Edge-FL, CCBS, and SEC

D. Model Training Time

The proposed Edge-FL framework also promoted faster training times than the two other approaches, the centralized traditional training used in CCBS and the stand-alone edge training utilized in SEC. Federated learning as a technique of decentralization reduced the computations involved and increased scalability.

TABLE IV. MODEL TRAINING TIME FOR CONTENT RECOMMENDATION

Model	Training Time
CCBS	320 s
SEC	280 s
Edge-FL	190 s

The average number of seconds needed to train the models for content recommendation is presented in the Table 4.

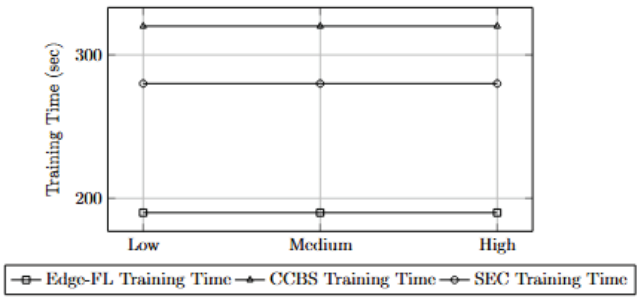

Fig. 5. Model Training Time Comparison for Edge-FL, CCBS, and SEC Models

Through performance metrics, the experiment proved that the proposed Edge-FL had higher performance compared to CCBS and SEC. Thus, the outcomes confirm the

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appropriateness of employing the elements of edge computing technology, dynamic bandwidth allocation, and federated learning for the improvement of methods for video streaming services. Besides, this approach increases the overall enjoyment of the application by the end users, optimizes the use of resources required, and provides for future growth appropriately.

V. Conclusion

It could also be seen the proposed framework of edge computing, DBA algorithms, and federated learning has presented extraordinary enhancements in mitigating video streaming services optimization. It can therefore be concluded that the Edge-FL framework was noticeably more effective with latency in the 20, 35 and 55 ms range for low efficiency, medium efficiency, and high efficiency, respectively which is much lower compared to the latency obtained for CCBS framework of 95, 140, and 210ms, and 35, 50, and 90 ms for SEC framework. The analysis of efficiency of bandwidth utilization showed that the proposed framework of Edge-FL was more efficient than others under all three conditions of low load, medium and high load: 90%, 80% and 70% correspondingly against 65%, 55%, and 40% of CCBS and 75%, 65%, and 50% of SEC correspondingly. Similar to Edge-FL results, QoE scores for the three traffic scenarios were 4.8, 4.5, and 4.2 for Edge-FL, better than CCBS with a 3.2, 2.8, and 2.2, and SEC at 4.1, 3.8, and 3.3. Also, the Edge-FL framework brought the training time down to 190 sec while it took 320 sec for CCBS and 280 sec for SEC. These enhancements reaffirm the adaptability of Edge-FL framework in handling problems like latency, bandwidth distribution and user experience to make future application of video streaming scalable and efficient.

References

[1] S. A. Quadri and K. S. Katakdhond, “Suspicious Activity Detection Using Convolution Neural Network,” J. Pharm. Negat. Results, vol. 13, no. S01, pp. 1235–1245, 2022, doi: 10.47750/pnr.2022.13.s01.151.

[2] M. Ekmal, E. Quyyum, M. Haris, and L. Abdullah, Proceedings of the Multimedia University Engineering Conference (MECON 2022). Atlantis Press International BV, 2023. doi: 10.2991/978-94-6463-082-4.

[3] T. J. Nandhini and K. Thinakaran, “An Enhanced Forensic Analysis and Security Surveillance Using Deep Reinforcement Learning,” 2023 2nd Int. Conf. Smart Technol. Smart Nation, SmartTechCon 2023, pp. 361–365, 2023, doi: 10.1109/SmartTechCon57526.2023.10391428.

[4] P. Arun Gawade, M. Kumar, and P. Balaramudu, “Real Time Efficient Foreground Extraction with Video Processing,” 2017 Int. Conf. Comput. Commun. Control Autom. ICCUBEA 2017, vol. 9, no. 1, pp. 65–70, 2018, doi: 10.1109/ICCUBEA.2017.8463957.

[5] T. Zhang, S. Liu, C. Xu, and H. Lu, “Mining semantic context information for intelligent video surveillance of traffic scenes,” IEEE Trans. Ind. Informatics, vol. 9, no. 1, pp. 149–160, 2013, doi: 10.1109/TII.2012.2218251.

[6] V. Dilli Ganesh and R. M. Bommi, “‘Prediction of Tool Wear by Using RGB Techniques in Comparison with Experimental Analysis,’” 2022 Int. Conf. Data Sci. Agents Artif. Intell. (ICDSAAI), Chennai, India, 2022, pp., 2022.

[7] M. S. Saravanan, D. Suresh, I. Sudha, T. J. Nandhini, S. Sivashankar, and K. Kalyani, “Quality Enhancement for Uninvited Content of Social Media Using Support Vector Machine and Alexnet,” 2023 Int. Conf. Data Sci. Agents Artif. Intell. ICDSAAI 2023, pp. 1–5, 2023, doi: 10.1109/ICDSAAI59313.2023.10452586.

[8] D. Devasenapathy, M. Raja, R. K. Dwibedi, N. Vinoth, T. Jayasudha, and V. D. Ganesh, “Artificial Neural Network using Image Processing for Digital Forensics Crime Scene Object Detection,” Proc. 2nd Int. Conf. Edge Comput. Appl. ICECAA 2023, no. Icecaa, pp. 652–656, 2023, doi: 10.1109/ICECAA58104.2023.10212302.

[9] L. Wang, L. Wang, M. Wen, Q. Zhuo, and W. Wang, “Background subtraction using incremental subspace learning,” Proc. - Int. Conf. Image Process. ICIP, vol. 5, no. January, 2007, doi: 10.1109/ICIP.2007.4379761.

[10] R. K. Varshney, A. Katiyar and P. Johri, "A Deep Learning Framework for Classifying Autism Spectrum Disorder from fMRI Images," 2024 International Conference on Cybernation and Computation (CYBERCOM), Dehradun, India, 2024, pp. 550-555, doi: 10.1109/CYBERCOM63683.2024.10803161

[11] Rastogi, D., Johri, P., Donelli, M., Kumar, L., Bindewari, S., Raghav, A., & Khatri, S. K. (2025). Brain Tumor Detection and Prediction in MRI Images Utilizing a Fine-Tuned Transfer Learning Model Integrated Within Deep Learning Frameworks. Life, 15(3), 327. https://doi.org/10.3390/life15030327

[12] T. J. Nandhini and K. Thinakaran, “Optimizing Forensic Investigation and Security Surveillance with Deep Reinforcement Learning Techniques,” 2023 Int. Conf. Data Sci. Agents Artif. Intell., pp. 1–5, 2023, doi: 10.1109/ICDSAAI59313.2023.10452551.

[13] D. Conte, P. Foggia, G. Percannella, F. Tufano, and M. Vento, “An experimental evaluation of foreground detection algorithms in real scenes,” EURASIP J. Adv. Signal Process., vol. 2010, no. ii, 2010, doi: 10.1155/2010/373941.

[14] C. Kim and J. N. Hwang, “Fast and automatic video object segmentation and tracking for content-based applications,” IEEE Trans. Circuits Syst. Video Technol., vol. 12, no. 2, pp. 122–129, 2002, doi: 10.1109/76.988659.

[15] A. Rana, P. Rawat, S. Vats, and V. Sharma, “Heatmap-Based Deep Learning Model for Network Attacks Classification,” SN Computer Science 2024 5:8, vol. 5, no. 8, pp. 1–12, Nov. 2024, doi: 10.1007/S42979-024-03447-3.

[16] Dhoundiyal, P., Sharma, V., Vats, S., & Rawat, P. (2025). A Progressive Hierarchical Model for Plant Disease Diagnosis. SN Computer Science, 6(2), 1-13.

[17] R. Vadisetty and A. Polamarasetti, "AI-Augmented Skill Development Roadmaps: Tailoring 12-Month Learning Paths for Future-Ready Careers in Education 4.0 and Industry 4.0," 2024 13th International Conference on System Modeling & Advancement in Research Trends (SMART), Moradabad, India, 2024, pp. 655-661, doi: 10.1109/SMART63812.2024.10882483.

[18] R. Vadisetty and A. Polamarasetti, "Gen AI for Real- Time Traffic Prediction and Autoscaling in Cloud Computing Education 4.0," 2024 13th International Conference on System Modeling & Advancement in Research Trends (SMART), Moradabad, India, 2024, pp. 735-741, doi: 10.1109/SMART63812.2024.10882511

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